

PAPER_686: UQFF Modulation for M87*: Shadow, Jet Power, and Ring Brightness

Author: Daniel Murphy

Framework: Unified Quantum Field Framework (UQFF) v5.30

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Class: UQFFModulationForM87 | CP4 #270

Source: grok_share_fc21e30c24b4.txt

Domain: Black Hole Imaging / M87

Abstract

M87* ($6.5 \times 10^9 M_\odot$), imaged by the Event Horizon Telescope in 2019, provides the largest SMBH test bed. The UQFF predicts: modified shadow radius $r_{\text{sh,UQFF}} = r_{\text{sh}} \sqrt{1 + f_{\text{TRZ}} \rho_{\text{UA}} / \rho_{\text{SCm}}}$, enhanced jet power $P_{\text{jet,UQFF}} = P_{\text{BZ}}(1 + f_{\text{TRZ}}) \sqrt{\rho_{\text{UA}} / \rho_{\text{SCm}}}$, and ring brightness ratio $(\rho_{\text{UA}} / \rho_{\text{SCm}})^{f_{\text{TRZ}}/4} \approx 1.58$.

Primary UQFF Equation

$$r_{\text{sh,UQFF}} = 3\sqrt{3} \frac{GM}{c^2} \sqrt{1 + f_{\text{TRZ}} \frac{\rho_{\text{UA}}}{\rho_{\text{SCm}}}}$$

Parameters

Parameter	Value	Description
$M_{\text{M87*}}$	$6.5 \times 10^9 M_\odot$	SMBH mass
r_{shadow}	$\sim 5.5 r_s$	Shadow radius
P_{jet}	10^{37} W	Blandford-Znajek power

UQFF Constants

Constant	Value	Description
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	Universal Aether density
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	Schwarzschild condensate
f_{TRZ}	0.1	Time-reversal zone factor
κ	$5 \times 10^{-4} \text{ day}^{-1}$	UQFF calibration constant
[SSq]	0.57	Superstring quenching factor
μ_J	$3.38 \times 10^{23} \text{ J}\cdot\text{m}$	Magnetic string coupling
γ	$5 \times 10^{-5} / 86400 \text{ s}^{-1}$	Aether oscillation decay

C++ Implementation

```
#include "UQFFModulationForM87.h"
```

```
UQFFModulationForM87 obj;  
auto result = obj.compute_primary(...); // primary equation  
obj.simulate_over_M(M_start, M_end, dM, "output.csv");
```

Python CP4 Calculator

```
from CondensedPhysics2 import UQFFModulationForM87Calculator
```

```
calc = UQFFModulationForM87Calculator()  
result = calc.compute() # returns all UQFF quantities  
print(result)
```

References

Event Horizon Telescope 2019 (M87*); Blandford & Znajek 1977; UQFF Session 173

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